

Bates et al. 2021 (RCPS) — checkpoint + attestation worksheet

What I actually read

Closely: §§1–3.1, especially §§2.1–2.2 and Theorem 1

Skimmed / skipped: Most of §§3.2–5, plus the more detailed application sections and appendices

Q1 — what is actually guaranteed

→ my answer: **Basically, the method gives us a prediction set where the actual risk is at most α , with probability at least $1-\delta$. So α is the risk level we're willing to accept, while δ is how likely it is that the guarantee itself fails. They're different because one is about the amount of error and the other is about how confident we are in the guarantee.**

→ section + page in my copy: **Definition 1, §1, p. 2**

Q2 — the price of the guarantee

→ my answer: **For the basic guarantee, the calibration data needs to be i.i.d., and the calibration data and test data need to come from the same distribution. The loss also needs to behave monotonically as the prediction set changes — basically, making the set larger shouldn't make the loss worse. The loss being controlled also needs to be fixed rather than changing after calibration.**

→ section + page: **§2.1, p. 4; Theorem 1, §2.2, p. 5**

Q3 — how the threshold is actually chosen

→ my answer: **It basically tries different values of λ using the calibration data. For each one it estimates the risk and calculates an upper confidence bound for that risk. It then picks the smallest λ where the upper bound is below α for that point and the larger values after it.**

→ section + page: **§2.2, pp. 4–5; Eq. (4)**

Q4 — monotonicity / nesting

→ my answer: **The nesting is important because it means that as λ changes, the prediction sets only grow and the risk changes in a predictable direction. That lets the paper use the pointwise confidence bounds to make a guarantee about the λ that was actually chosen from the data. Without this monotonicity, the same argument doesn't work and we'd need a stronger uniform-convergence result instead.**

→ section + page: **§2.1–§2.2, pp. 4–5; Theorem 1**

Q5 — mapping to OUR design

→ my answer:

(a) The loss would be the pre-defined CAL-RISK loss we're using to measure an unacceptable error in the system.

(b) No, I don't think our calibration data satisfies the assumption directly. Our data are clustered, with tasks nested within origins/publishers, and they're design-weighted. That's different from the i.i.d. calibration setup in Bates et al., so we can't just claim their finite-sample guarantee applies to our data.

(c) CA-01.A4 says we should therefore report the result under our actual clustered/design-weighted study design, rather than claiming that the Bates finite-sample RCPS guarantee has been established.

Attestation row

source: Bates, S., Angelopoulos, A., Lei, L., Malik, J., & Jordan, M. I. (2021). *Distribution-free, risk-controlling prediction sets*. Journal of the ACM, 68(6), Article 43. DOI 10.1145/3478535.

sections/pages read: §§1–3.1, especially §§2.1–2.2, pp. 4–5, plus selected appendix material

claim supported: **The RCPS procedure can control risk at level α with probability at least $1-\delta$, provided the assumptions needed for the procedure are satisfied.**

assumptions/limits: **The main result assumes i.i.d. calibration data, the calibration and test data coming from the same distribution, a monotone loss, and nested prediction sets, along with a valid pointwise upper confidence bound. Without the monotonicity, the paper says a uniform-convergence argument would be needed. Our clustered and design-weighted data therefore don't automatically satisfy the basic finite-sample guarantee.**

decision affected: **CAL-RISK partition (F-03/F-04); risk-control procedure in F-08; CA-01.A4 validity conditions; H-P risk gate**

date / reader: **2026-08-10 / Fahmi Alshahabi**

reviewer spot-check: — pending —